

Research article

Comparing total suspended solids sensors for improved dewatering of wastewater from non-sewered sanitation

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ABSTRACT

Almost half of the global population relies on non-sewered sanitation, wherein wastewater is treated directly at source or stored onsite and transported by road to treatment facilities (i.e., fecal sludge management). For the protection of public health and environmental ecosystems, efficient non-sewered sanitation systems are needed, in particular in urban areas where infrastructure lags behind rapid urbanization. Wastewater delivered to treatment plants in non-sewered areas has highly variable characteristics, as it arrives in batches from variable infrastructure. Established systems to handle this variability are often land- or energy-intensive, while operation of low-footprint alternatives strongly depends on the wastewater characteristics, making them difficult to implement for highly variable wastewater. The use of sensors could improve process control and increase the robustness of treatment technologies, while reducing the required footprint. However, there are no studies comparing the effectiveness of different sensor methods to measure solids in non-sewered wastewater. We therefore chose six sensors for predicting solids in wastewater and evaluated their performance in variable wastewater with a range of solids content, inorganic particles, salts, and fats. This work demonstrated that the sensors were affected differently by the variable wastewater. Micro-wave and Coriolis-flow technologies were strongly impacted by the addition of fat, oil, and grease as well as salts and inorganics, whereas optical sensors were mostly affected by inorganics, with optical light-attenuation having the lowest overall error, making it the best option for variable wastewater. A low-cost optical light-attenuation sensor had similar accuracy as a more expensive one when the solids content was low, indicating its value for inexpensive monitoring of effluent solids in decentralized settings. Additionally, three sensors were applied for process control of chemically enhanced dewatering with conditioners and geotextiles. Controlling conditioner dosing with sensors in the flocculation tank improved performance with 77% solids reduction, which is in the range of the alternative, time-intensive jar-test method (83% reduction). This study reveals that sensors can be useful in non-sewered wastewater management when the type of input wastewater is accounted for and that even low-cost sensors (<100 USD) can be suitable for many applications.

1. Introduction

The health of any community directly depends on the appropriate treatment of wastewater. More than one million deaths worldwide could be prevented yearly with safe water, sanitation and hygiene (Wolf et al., 2023). While about half of the global population is connected to sewer systems that can transport wastewater to treatment facilities, the rest relies on non-sewered sanitation systems that treat wastewater onsite or

transport by road to a treatment plant (UNICEF/WHO, 2023). Non-sewered sanitation systems can provide adequate sanitation if the entire service chain is functioning. However, the composition of wastewater from non-sewered sanitation systems is highly variable, as it is generated from variable infrastructure and arrives in batches at treatment facilities (Strande, 2024). This makes it difficult to set standard treatment values and adequately treat the wastewater. Wastewater is defined here as blackwater and greywater. Blackwater is excreta (i.e.

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feces and urine), flush water, and cleaning material. Greywater is wastewater from bathing, cooking, or cleaning. This research is relevant for the treatment of blackwater close to or at source of production and treatment of stored wastewater transported by road to treatment, commonly referred to as “fecal sludge” (Strande et al., 2023).

When wastewater from non-sewered sanitation arrives at treatment facilities, it is often first dewatered to remove the unbound water and enable complete treatment of the resulting solid and liquid stream. Established robust technologies for dewatering, such as drying beds, are land- or energy-intensive. The efficiency of low-footprint alternatives, such as chemically enhanced dewatering through the addition of conditioners (coagulants and flocculants) (Gold et al., 2016), is reduced through over- and under-dosing. Reliable and robust conditioner dosage can improve the efficiency of low-footprint treatment, which is required to accommodate rapid urbanization and population growth.

The conditioner dosage used in centralized wastewater treatment is often based on measurements of the amount of total suspended solids (TSS) or total solids (TS). In wastewater from onsite storage containments, TSS was predicted by optical sensors (commonly referred to as turbidity sensors) (Junglen et al., 2020). Additionally, TS was predicted by texture and color (Ward et al., 2021). To control conditioner dosing in variable environments, sensors measuring solid particles in a liquid can provide real-time measurements of TSS and TS and allow for appropriate conditioner adjustments. Established and tested technologies for this include optical (light-scattering and light-attenuation) (Mels et al., 2004; Shaw et al., 2022), Coriolis-flow, and micro-wave sensors. Optical light-scattering sensors measure the scattering of light by particles (DIN, 2016), while light-attenuation sensors measure attenuation of light by the liquid, which account for either absorbed or scattered light (Davies-Colley and Smith, 2001). The Coriolis flow principle quantifies the mass flow and density of a fluid by measuring the excitation frequency induced by the mass flow on a movable tube (Anklin et al., 2006). Micro-wave sensors predict solid content by measuring the permittivity of the fluid (Yuan and Pal, 1995). One additional sensing technology that has not yet been used for wastewater measurements but should be effective is ultra-sonic sensing, which measures dispersion via ultra-sonic wave (Weser et al., 2014). Although there are many examples of sensors capable of measuring solids in wastewater, it is unclear how well these sensors function under varying wastewater characteristics from non-sewered sanitation.

Studies into sensor robustness under varying conditions are needed, as the characteristics of the wastewater strongly influence the TSS prediction of these sensor technologies. For example, optical sensors (light-scattering and light-attenuation) depend on the size, shape, and color of the particle (Davies-Colley and Smith, 2001). Coriolis-flow sensors depend on the density of particles and the contribution of the dissolved fraction to the overall density. Micro-wave method depends on the permittivity of the material, and the ultrasonic method on the dispersion, which both can be influenced by dissolved and undissolved matter (Mels et al., 2004; Weser et al., 2014). The wastewater from non-sewered sanitation can be as dilute as municipal wastewater (<1 g TS/L) up to slurry fluid (>50 g TS/L) (Velkushanova et al., 2021), depending on the type of infrastructure, storage and transport. In addition, though TSS is a good indicator for micro-wave, ultra-sonic, and optical-light sensors, it is expected that dissolved solids can interfere with the measurement. This high matrix dependency makes the use of a single calibration for different wastewater types difficult or even impossible. As exists for other more well established uses of sensors (Sambito and Freni, 2021), it would be useful to build up enough knowledge to have models to select sensors for non-sewered sanitation systems.

The objective of this study was to validate six types of sensors for a range of solids observed in non-sewered sanitation, in order to aid in sensor selection for different process flow configurations and the dosing of conditioners. Six sensors were evaluated in a laboratory pilot-scale test at Eawag in Switzerland, and three sensors were then field-tested

in a pilot-scale implementation of a process flow with conditioner dosing at CubeX in Lebanon. For evaluating sensor performance, we mainly used TSS, and in some cases, TS. The full data set reporting both TSS and TS is available in the supplementary information (SI). Results of this study are directly applicable for process control and monitoring of performance of blackwater treatment at scales from in-building to semi-centralized (e.g., fecal sludge management). The results can be transferred to other types of wastewater, such as municipal or industrial wastewater or landfill leachate. This study presents a guidance for sensor selection to predict and monitor TSS in non-sewered sanitation systems.

2. Material and methods

2.1. Sensors

As presented in Table 1, six different sensors were validated to predict the TSS of blackwater in a pilot installation at Eawag in Switzerland, here after referred to as the “TSS sensor pilot”. A picture of the TSS sensor pilot, including all six sensors, is provided in S1. Three of these sensors were then validated for conditioner dosing for improved dewatering of stored wastewater (“fecal sludge”) in Beirut, Lebanon, hereafter referred to as the “dewatering pilot”. The sensors use five different principles of measurement and are commercially available for use in wastewater treatment or the food industry. For all sensors, the default factory settings were selected. For CUS51D, the 90° light-scattering factory model was selected. For Valmet TS measurements, a single-point calibration was required, which was done at 2.5 gTSS/L with blackwater.

Promass Q300 measures density to predict TS, as density is affected by total dissolved solids and TSS. As temperature directly affects the density of water, the measurements from Promass Q300 were corrected for temperature during data analysis by adding a correction factor of 0.0002525 kg/m³ per degree Celsius deviation from 18 °C. The correction factor was determined by measuring the density of blackwater with Promass Q300 at two temperatures (18 and 19.5 °C). The values of the other sensors were not temperature corrected, as they either have internal temperature correction or temperature has a negligible impact on the sensor reading.

Data from TSW-10 was retrieved with the microcontroller Arduino Micro (5 V output) in the TSS sensor pilot and with the microcontroller LilyGo TTGO LoRa32 (4.2 V output) in the dewatering pilot. To compare the analog output signals of both microcontrollers, the analog output signal of LilyGo TTGO LoRa32 was corrected following the signal-turbidity curve on the TSW-10 datasheet (Amphenol-Corporation, 2019) based on the difference of the analog output signal between the two microcontrollers when measuring in deionized water at room temperature.

2.2. Pilot installations

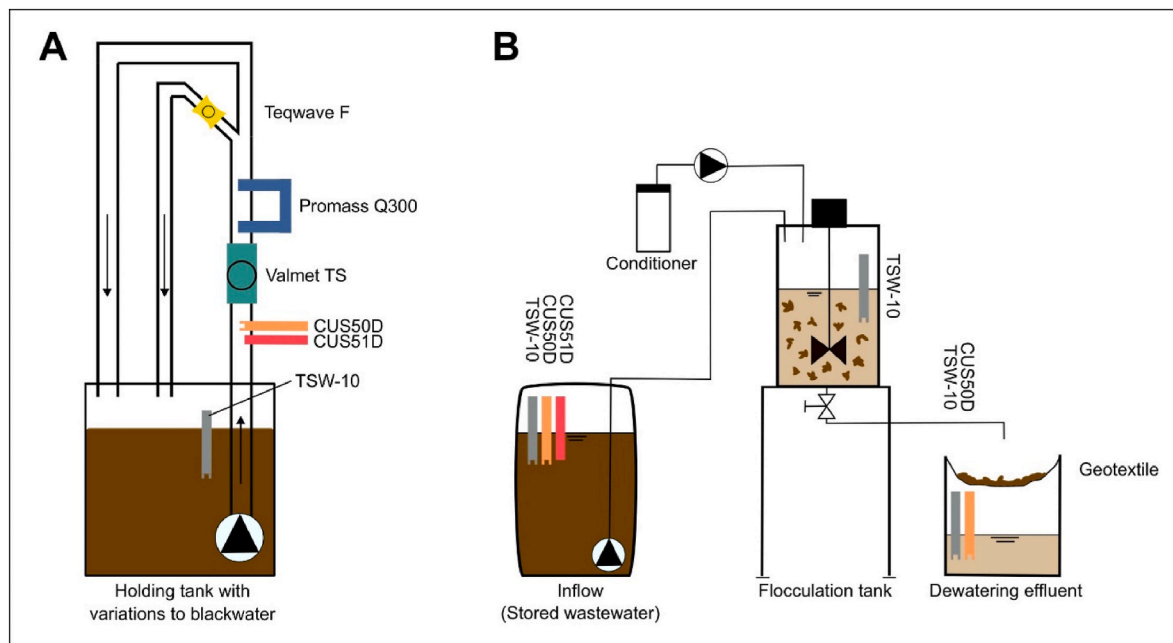
The TSS sensor pilot (Fig. 1A) was designed to evaluate the six sensors via a direct comparison in the same flow of blackwater. Controlled trials were run to represent different ranges of blackwater characteristics. For parallel testing of sensors in each trial, the holding tank was filled with a blackwater variation, which was then pumped with a KSB Amarex S50-160 pump through the pipe network. Blackwater variations were homogenized directly in the container by maintaining a high pumping rate, which had the added benefit of minimizing the risk of sensor fouling by inducing high shear rates at the sensor surface.

Based on the results of the TSS sensor pilot, three optical sensors (CUS51D, CUS50D, and TSW-10) were selected for additional testing at the dewatering pilot (Fig. 1B) at three different locations in a flocculation and dewatering process flow: (1) inflow of stored wastewater (“fecal sludge”), (2) flocculation tank, and (3) dewatered effluent. Stored wastewater was pumped with a submersible pump (V250f) into

Table 1

Overview of the evaluated and validated sensors on the two different pilot installations.

Sensor	Manufacturer	Principle	Measurement parameter and factory models	Additional recorded parameter	TSS sensor pilot	Dewatering pilot
Valmet TS	Valmet	Micro-wave	TS [g/L] Factory model calibration at 2 gTSS/L with blackwater, no conductivity correction	Conductivity [mS/cm]	X	
Promass Q300	Endress + Hauser	Coriolis-flow	Density [kg/L], temperature correction	Temperature [°C], Mass flow [kg/s]	X	
Teqwave F	Endress + Hauser	Ultra-sonic dispersion	Dispersion coefficient [-]	Speed of sound [m/S], Density [kg/m]	X	
CUS51D	Endress + Hauser	Optical light-scattering	90° light-scattering [FNU] Factory model		X	X
CUS50D	Endress + Hauser	Optical light-attenuation	Light-attenuation at width 5 mm [AU]	Attenuation at width 10 mm [AU]	X	X
TSW-10	Amphenol	Optical light-attenuation	Light-attenuation analog output signal [V]		X	X

**Fig. 1.** (A) Total suspended solids (TSS) sensor pilot in which all six sensors were tested. (B) Dewatering pilot in which only CUS50, CUS51D, and TSW-10 were tested at the indicated locations.

the flocculation tank, which had a volume of 100 L and was equipped with a 2-blade flat paddle impeller (35 cm × 15 cm). Conditioner (polyacrylamide, OTE PK 1350) was pumped directly into the flocculation tank using a peristaltic dosing pump (Welco WP1100). The flocculation tank contained a bottom outlet that was opened manually with a valve to allow conditioned wastewater to flow onto the dewatering geotextile (Stratex M 300), which had 0.3 mm pore size and was mounted over a 100-L barrel.

2.3. Blackwater and stored wastewater

To simulate the properties of blackwater and stored wastewater (“fecal sludge”) that might influence the measurements, different variations of blackwater were prepared and are summarized in Table 2. All variations were used in the TSS sensor pilot, whereas only stored wastewater was used in the dewatering pilot.

Blackwater was collected in the basement of the NEST building (Eawag, 2023), directly from the pipes that convey blackwater from the toilets. The NEST building has living, recreational and office space with installed urine-separating (Laufen Safe toilets) and mixed toilets. Feces were collected from a dry toilet at the Eawag office building in

Dübendorf, Switzerland. Feces were stored at −18 °C for less than three months and were thawed at room temperature before the trials. Toilet paper was used in a ratio of 30 sheets of toilet paper to 1 kg of feces. No urine was added to the blackwater to avoid additional dissolved solids that could affect the sensor readings. To test variations in blackwater, additives included peanut oil to represent fat, oil, and grease (FOG); NaCl to represent salts and dissolved solids; NaOH and HCl to control the pH; and commercially available potter’s clay, kaolin, and vermiculite for particle size distribution. To provide a direct comparison, activated sludge was evaluated with the sensors, as they are already widely used in municipal wastewater treatment.

For trials in the conditioner pilot, stored wastewater was collected from ten different containments in Aintoura, Lebanon, referred to as Leb-1 to 9 in this study. Additionally, stored wastewater from four containments in Kampala, Uganda, was used for additional validation of the TSS prediction.

2.4. Experiments

In the TSS sensor pilot, the sensors were validated in a two-step procedure. First, a TSS prediction curve was determined for each

Table 2

Overview of the various blackwater variations generated for the trials.

Trial name	Location	Target property	Variations to blackwater	Target TSS
Calibration	TSS sensor pilot	Total suspended solids (TSS)	Controlled blackwater recipe for calibration: feces and toilet paper were added step-wise to tap water to achieve the indicated TSS range	0–27.2 g/L
Solids variability	TSS sensor pilot	Variable (increasing and decreasing) TSS	Stepwise addition of tap water, feces, and toilet paper to increase or decrease TSS	1.9–11.6 g/L
FOG	TSS sensor pilot	Fat, oil, and grease	Blackwater with stepwise addition of peanut oil	2.0 g/L
pH	TSS sensor pilot	pH	Blackwater with stepwise addition of NaOH or HCl to a pH range of 5.0–9.0	2.2 g/L
Salt	TSS sensor pilot	Salts, total dissolved solids	Blackwater with stepwise addition of NaCl to an electrical conductivity range of 1.5–30.4 mS/cm	2.0 g/L
PSD1 (Particle size distribution)	TSS sensor pilot	Change in particle size distribution and particle properties	3 combinations in 2 concentrations each - Tap water and kaolin - Tap water and potter's clay - Tap water and vermiculite	2.0 g/L and 3.7 g/L
PSD2 (Particle size distribution)	TSS sensor pilot	Change in particle size distribution and particle properties	Different mixtures of blackwater (60–80%), kaolin (7.5–25%), potter's clay (7.5–25%), vermiculite (5–25%)	2.5–3.1 g/L
PSD3 (Particle size distribution)	TSS sensor pilot	Change in particle size distribution and particle properties	Tap water with stepwise addition of feces, toilet paper, kaolin, potter's clay, vermiculite, NaCl, and peanut oil	3.3–11.3 g/L
Activated sludge	TSS sensor pilot	Change in sludge type	Activated sludge diluted stepwise with tap water	1.4–3.0 g/L
Stored wastewater (Leb-1 to 10)	Dewatering pilot	Change in wastewater type	10 collections from a range of containments in Aintoura, Lebanon and informal settlements around Baalbek	1.1–15.7 g/L
Stored wastewater	Dewatering pilot (inflow only)	Change in wastewater type	4 collections from a range of containments collected in Kampala, Uganda	0.4–11.0 g/L

sensor via the calibration of blackwater (Table 2). Second, the calibration was validated with blackwater with a range of different ingredients, as summarized in Table 2. Prior to each trial, the blackwater variation was pumped through the system for 7–10 min to ensure homogenization, temperature adaption, and stabilization of sensor readings. Then, for each blackwater variant, the comprising ingredients were added step-by-step, with 7–10 min of pumping between each step. Five minutes after each addition, the next 2 min of data was retrieved and averaged for further data analysis. Grab samples were taken from the holding tank for laboratory analysis. During the experiments, water temperature was monitored with the integrated temperature sensor of the Promass Q300 and was maintained between 17 and 23 °C.

In the dewatering pilot, the CUS50D, CUS51D, and TSW-10 sensors were applied to a dewatering process flow. For this, the sensors were tested in the inflow (stored wastewater) to validate the TSS prediction and optimal dosage of conditioner, in the flocculation tank to evaluate real-time control of conditioner dosage, and in the effluent to validate effluent monitoring of TSS. For each trial, stored wastewater was pumped into the flocculation tank. The conditioner was dosed directly into the flocculation tank and mixed. A barrel was used to collect the effluent after geotextile dewatering. Grab samples were taken of the inflow stored wastewater before the experiment and of the dewatering effluent after geotextile for laboratory analysis. The optimal conditioner dosage was determined using a jar-test as described in (Shaw et al., 2022), wherein different dosages were added to five jars of the same stored wastewater. The optimal dosage was defined as the jar with lowest turbidity measured in the supernatant after 5 min of settling. This conditioner volume was then added stepwise into the flocculation tank in 10–16 additions with one addition per minute. To evaluate the conditioner dosing in real time, sensor data from the flocculation tank was recorded during the trials. Based on this data, a second set of trials was conducted wherein the dosage of conditioner was not based on jar-test predictions and was instead controlled only by the TSW-10 sensor in the flocculation tank. For this set of experiments, composite samples of Leb-1 to 9 were used, labelled in this study as Auto-1 to 5. To evaluate the effluent monitoring of TSS, the sensor data was compared with laboratory data.

2.5. Laboratory analysis

The analysis of turbidity, TSS, VSS (volatile suspended solids), TS

and COD (chemical oxygen demand) was conducted following standard methods (Apha, 1985; Velkushanova et al., 2021). The analysis of pH and EC was conducted using a portable measuring device (Multi 3320 WTW). For assessing the color photographs of 10-mL aliquots of blackwater were analyzed using the methodology from (Ward et al., 2021). Particle size distribution was determined through gravimetric analysis using serial filtration (Metcalf and Eddy, 2003).

2.6. Data analysis

Sensor calibration curves were chosen by comparing the correlation coefficient (r^2) of a linear, exponential, and polynomial (2nd and 3rd degree) model. The curve with the lowest r^2 was selected. The coefficient of determination (R^2) and root-mean-square error (RMSE) were calculated to evaluate the prediction of TSS by these curves.

2.7. Quality assurance & quality control

For the TSS sensor pilot, all TSS and TS samples were measured in duplicate. If the measurements deviated by more than 10% between two samples, the measurements were repeated. Before every experiment, the kaolin standard solution (2 g/L) was run through the sensor to detect any sensor fouling. For the dewatering pilot, 10% of all TSS measurements where conducted in triplicate. Additionally, wastewater Leb-1 was run in triplicate, Leb-3 in duplicate through the entire pilot system. Any observed uncertainties are discussed in results and discussion section.

3. Results and discussion

3.1. Controlled blackwater experiments at Eawag

The TSS sensor pilot (Fig. 1A) conducted at Eawag in Switzerland was designed to evaluate the six sensors in the same stream of blackwater, allowing for a direct comparison between the sensors. With this sensor set-up, we ran controlled trials representing different ranges of blackwater characteristics. Sensor readings and laboratory TSS measurements for all trials listed in Table 2 are presented in Fig. 2. Calibration curves were calculated using the sensor and laboratory data from the blackwater calibration trial and are shown as a black line with grey shading indicating a 10% deviation from this curve (see SI for

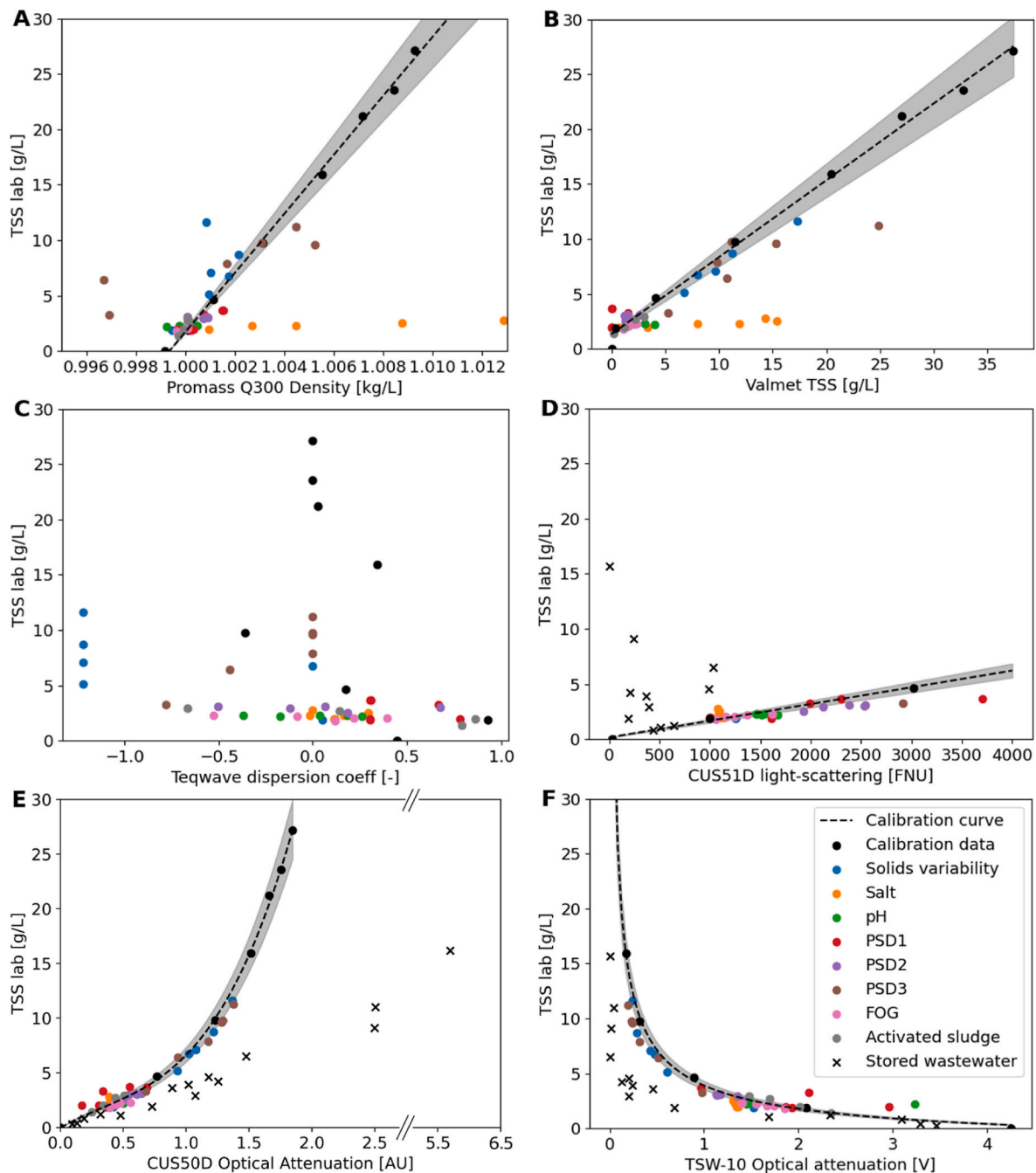


Fig. 2. Controlled blackwater experiments at Eawag. Scatter plots of all sensor measurements from conducted experiments in relation to total suspended solids (TSS) for each sensor: (A) Promass Q300, (B) Valmet TSS, (C) Teqwave F, (D) CUS51D light-scattering, (E) CUS50D optical attenuation, and (F) TSW-10 optical attenuation. Model fit according to Table S3. Stored wastewater is only shown for TSW-10, CUS50D, and CUS51D (D–F). Grey shading depicts a 10% deviation from calibration curve.

calibration functions). The results from modified blackwater samples reflecting the expected variety in wastewater for each sensor were compared to the calibration curve. In the following section, the results for each sensor are presented along with a discussion of possible applications.

3.1.1. Promass Q300 (coriolis-flow)

As a Coriolis-flow sensor, the Promass Q300 was expected to show a clear correlation to TSS when the material composition remains constant, in this case water, toilet paper and feces. This fits with the fairly accurate calibration curve ($R^2 = 0.997$), though the sensor showed an expected deviation from this curve upon the addition of salt or high

densities of particles. For example, a change from 0.95 mS/cm to 1.28 mS/cm increased the sensor reading by 50% from the laboratory-measured value of 2 gTSS/L. Calibrating the sensors with TS rather than TSS improved the results, but the overall error was still high (RMSE = 5.6 g TS/L). Although this sensor is applicable in larger, centralized wastewater treatment facilities, the high costs and sensitivity to salts may rule it out for small-scale blackwater treatment systems.

3.1.2. Valmet TS (microwave)

Valmet TS had a strong linear calibration curve ($R^2 = 0.995$), with a few blackwater variations that deviated from this. The Valmet TS microwave sensor was expected to react most strongly to any compounds

affecting the permittivity of liquids (Ellison et al., 1998; Spohner, 2017). The data from the sensor supported this, with the addition of salts and oils causing the strongest deviation from the linear calibration curve ($R^2 = 0.995$). To better show the effect of salts on the permittivity, automatic electrical conductivity compensation provided by Valmet TS was deactivated for this trial. It is therefore assumed, that the effect of salts addition can be adequately compensated if electrical conductivity compensation is activated. However, the addition of only a small amount of vegetable oil (22 mL into 100 L) increased the Valmet TS sensor measurement by 20%. Similar to the Promass Q300 sensor, calibration with TS instead of TSS reduced the error (RMSE = 1.7 gTS/L). This is because the high sensitivity to salts is less important for prediction of TS, as TS includes dissolved solids (e.g., salts). These sensor operating principles may also explain the weaker but still notable impact of the change in particle composition (PSD) experiments. As such, this sensor might work well when the variability in the wastewater composition is low (e.g. constant ratio of FOG and electrical conductivity to solids), though there is a high discrepancy in the TSS values (>5 gTSS/L difference). This might include fresh blackwater directly from urine-diverted toilets, where the main contributions to the wastewater are flush water, cleansing material, and feces. An application with variable urine content could be possible with adequate compensation for the electrical conductivity.

3.1.3. Teqwave F (ultra-sonic dispersion)

It was not possible to fit a calibration curve for Teqwave F. This likely indicates that this technology is not suitable for measuring the heterogenic particle characteristics in blackwater, as the scattering behavior of ultrasonic waves depends on the material properties (Weser et al., 2014). Though wastewater treatment is not the intended field of application of Teqwave F, other studies have shown that ultrasonic measurement had the potential to estimate solids in water (Pallarès et al., 2021; Kuang et al., 2023). Thus, it might be that other devices besides the Teqwave F that are based on the same principle might work to measure solids in blackwater. As such, further research would be needed to definitively confirm the non-viability of ultrasonic dispersion readings in wastewater, though our initial studies here seem to indicate that at least the Teqwave F is not suitable for these purposes.

3.1.4. CUS51D (light-scattering)

Because the measurement principle of CUS51D (90° FNU model, 90°-light-scattered method) is only valid within the range of <5 g/L TSS (Endress+Hauser, 2023), this sensor was calibrated only in this range ($R^2 = 0.992$). As light-scattering measurements are influenced by particulate properties like shape and size (Kitchen et al., 2017), the accuracy was most affected by PSD variations, which contained increasing amounts of fine inorganic particles that scatter more light than organic ones (Bright et al., 2018). Kaolin and potter's clay contained mostly smaller inorganic particles, with median diameters of 18 µm and 5 µm respectively, whereas blackwater contains significant organic material from feces and had a medium particle diameter of 164 µm in this study. This explains why measurements in trials with kaolin and potter's clay additions (PSD experiment) led to an overestimation of the TSS. The variable sensor readings for stored wastewater ("fecal sludge") can be for similar reasons, as there are large variations in particle size distribution (median diameter from 8 to 184 µm) (Ward et al., 2023) and fraction of organic particles (VSS/TSS in Table S4 in the SI). Overall, these results suggest that CUS51D with 90° FNU model is appropriate for blackwater with low TSS values (<5 g/L) and with a low variation in particle composition. This could apply to, for instance, blackwater from flush toilets without storage or additional inorganic inputs.

For measuring higher TSS values in the future, testing could be conducted on the additional factory model of the CUS51D that employs a 135°-light-scattering method. To quickly test this theory using this sensor, preliminary measurements in stored wastewater ("Excess sludge", Fig. S5 in the SI) showed a $R^2 = 0.91$ in a TSS range of 1.1–15.7

g/L. This is supported by recent work that found a relationship between TSS and turbidity (NTU) in stored wastewater in a range of 10–80 gTSS/L ($R^2 = 0.99$) (Junglen et al., 2020). For this study, details as to why the 135°-light-scattering method provides better insights into higher TSS values has not been investigated, as this requires insights into proprietary software of the models.

3.1.5. CUS50D (light attenuation)

Of all tested sensors, CUS50D generated the best prediction ($R^2 = 0.999$) and best validation (RMSE = 0.7 gTSS/L). In contrast to the Promass Q300 and Valmet TS sensors, calibration with TS did not improve the validation (RMSE = 3.3 gTSS/L), as salt does not affect light attenuation as strongly as it contributes to TS. Similar to the CUS51D light-scattering sensor, the strongest, but small, deviation from the calibration here was observed with the PSD trials, which can be explained by the dependence of light attenuation (scattering and absorption of light) on particle characteristics. In fact, light attenuation is known to be increased by fine and inorganic particles that increase the refractive index (Bright et al., 2018; Davies-Colley et al., 2014; Davies-Colley and Smith, 2001) without changing the TSS. A higher uncertainty is therefore expected for measurements in stored wastewater, where fluctuations in VSS/TSS are larger (Table S4). A clear trend for this cannot be seen from the blackwater trials, however there is an overall higher attenuation of the readings in stored wastewater (Fig. 2E). The attenuation values were so high as to prevent measurements of the visual clarity (Davies-Colley et al., 2014), which was instead analyzed via pictures. As darker colors absorb more visible light, the overall darker color observed for stored wastewater than for blackwater (full data set in Table S4) may account for the higher light attenuation of stored wastewater. Within the stored wastewater samples, Leb-8 had the darkest color and highest attenuation. This suggests that readings of color combined with light attenuation might increase the accuracy of TSS predictions of stored wastewater. However, Leb-2 and Leb-5 both also had a rather dark color but no clear difference in attenuation to the rest of the stored wastewater (Fig. S5). Though previous work suggested that simple measurement principles can predict stored wastewater characteristics, such as using color and texture to predict TS (Ward et al., 2021). Regardless, even though this data does not explain why the attenuation of Leb-8 is clearly higher, eight out of nine samples from one city still follow a similar attenuation pattern, which suggests that this sensor could be used to predict TSS in stored wastewater, though it may generate frequent outliers.

Prior to any application, however, more data is required to validate the performance of this sensor over the entire range of expected TSS in the application area. For instance, in this study, stored wastewater had a measured range of up to 15.7 gTSS/L, though many areas have reported a higher TSS, such as up to 49 gTSS/L in Dakar, Senegal (Ward et al., 2019). Even with this caveat, CUS50D best predicted the TSS over the entire validation range of all the tested sensors, indicating that it has the potential to be used at different points in a blackwater treatment system over a broad range of TSS, even in the presence of variable FOG, organic and inorganic compounds, and salt content.

3.1.6. TSW-10 (light attenuation)

Light-attenuation sensor TSW-10 was tested as a cost-effective alternative to CUS50D, with differences in the software, material, electronic components, and the distance between the emitter and the sensors. Overall, TSW-10 behaved similarly to CUS50D, with a good calibration ($R^2 = 0.997$) and a slightly weaker validation (RMSE = 1.1 gTSS/L). This fits well with previous studies that found similarly high R^2 values using TSW-10 to predict turbidity at low concentrations (>3 V output signal) in surface water and aquaculture applications (Hakim et al., 2019; Valenzuela et al., 2018). In this study, a low measurement resolution was found at higher TSS values of around 7 gTSS/L for blackwater and 5 gTSS/L for stored wastewater, as indicated by the increasing slope in Fig. 2F. This makes sense for a light-attenuation

sensor, as these high TSS concentrations permit only a small amount of light to pass from the light emitter to the receiver, which lowers the resolution.

However, TSW-10 did have a lower measurement range compared to CUS50D. This may be due to the longer distance between the light emitter and receiver, at 6 mm for TSW-10 compared to 5 mm for CUS50D, and different electronic and software components. Unfortunately, the TSW-10 is not suitable for applications in which the TSS is expected to be higher (>5 g/L), though sensors could be adapted to fit the concentration. In fact, several studies describe the development of low-cost turbidity sensors, mostly for surface waters (Droujko and Molnar, 2022; Gillett and Marchiori, 2019; Murphy et al., 2015), with the objective to increase accuracy at a low turbidity to fit the purpose of solids monitoring in surface water. Based on the experience of these authors, we suggest that low-cost sensor can similarly be adapted for better resolution at a higher TSS, such as by decreasing the distance between light emitter and receiver.

3.2. Application to process control in dewatering pilot study

Next, the findings from the above study were validated in a dewatering pilot study conducted at CubeX in Lebanon, where stored wastewater is flocculated and dewatered. This dewatering pilot (Fig. 1B) tested three optical sensors (CUS51D, CUS50D, and TSW-10) at three different locations of a flocculation and dewatering process flow: (1) inflow of stored wastewater (“fecal sludge”), (2) flocculation tank, and (3) dewatered effluent. The optical sensors were selected for this pilot because of their overall low RSME for the blackwater trials and the possibility to directly install them in a tank, whereas the other tested sensors need to be integrated into a piped system. The sensors were trialed at different locations in the process flow as indicated in Fig. 1B, with the goal of evaluating and validating their potential for predicting conditioner dosage, monitoring and controlling the flocculation process, and monitoring the effluent after dewatering.

3.2.1. Inflow measurement to predict conditioner dosage (CUS50D and TSW-10 light attenuation)

For real-time predictions of conditioner dosing, we next sought to determine the optimal dosage based on the sensor reading. The dosage of poly-acrylamide conditioner has been reported to correlate linearly with TSS in stored wastewater ($R^2 = 0.8$) (Shaw et al., 2022). Therefore, results from the TSS sensor were directly used to find the best prediction curve for required conditioner dosages determined by jar tests. For CUS50D, the TSS prediction for the blackwater variations followed a polynomial function, however, for the stored wastewater a linear trend appears to be more suitable on this range (Fig. 2E), therefore a linear

function was used. Two data points were removed from the calibration, namely Leb-6 (outlier) and Leb-8 (out of detection range). For TSW-10, 5 data points were removed from the calibration, namely Leb-6 (outlier) and Leb-1, 2, 5 and 8 (out of detection range), to give a regression to the same exponential function as for the blackwater calibration (Fig. 2F). The resulting regression curves are illustrated in Fig. 3.

It is important to consider that there is not one distinct dosage volume where flocculation occurs, but rather a dosage window where flocculation is sufficient for a further treatment step. Therefore, the grey error bars illustrate a dosage window of $\pm 38\%$, as a dosage variation of 38% can still reduce dewatering time by 85% as measured by capillary suction time (Shaw et al., 2022). Similarly, in another study, altering the dosage by $\pm 50\%$ did not change the filtration efficiency with geotextiles (Rhodes-Dicker et al., 2022). With this window, almost all the regression points are within the regression curve, and both regression curves indicate a clear correlation between the model and the dosages. The data suggests that both CUS50D and TSW-10 can be used to roughly predict conditioner dosage by considering the measuring range of the sensor. Further piloting is required to evaluate whether the decrease in treatment performance by the occasional under- or overdosing is justified by the time savings over jar testing.

3.2.2. Controlling conditioner dosing by monitoring flocculation process (TSW-10 light attenuation)

Another useful application of a sensor would be to directly measure TSS in the flocculation tank to monitor the flocculation process with step-wise addition of conditioners. To choose which sensor to use, we referred to the jar tests, which indicated that the TSS of the supernatant after settling was low for all samples (<10 mg/L). As this indicates that every sample of this study could theoretically reach a low light attenuation in the supernatant at the optimal dosage, a TSW-10 sensor was installed in the top 10 cm layer of the 50-L flocculation reactor. To reach the optimal dosage as determined above, conditioner was added step-wise and the TSW-10 signal output was monitored (Fig. 4). The flocculated wastewater was then dewatered using a geotextile. In the top layer of the flocculation tank, the output signal increased to at least 2.5 V in all samples, indicating a clear decrease in solids, with the majority higher than 3.8 V and thus close to that of drinking water (4.95 V). This trend held even for samples outside of the detection range of TSW-10, such as for Leb-2, 5, and 8. Though there was some fluctuation in the readings, this could be explained by observations of flocs sticking to the sensor surface (Leb-9) or poor flocculation (Leb-1-c). Additionally, while both Leb-5 and Leb-8 exhibited a comparably lower final output signal, the high initial TSS of these samples increased the amount of floating flocs that can obstruct the sensor and increase the measured turbidity, resulting in a lower output signal.

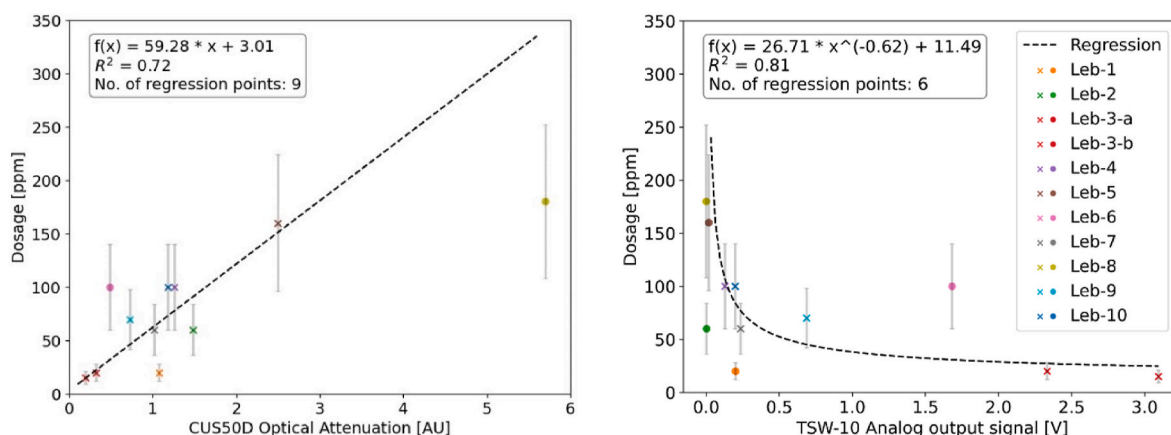


Fig. 3. Determining the optimal conditioner dosage in relation to the readings from the TSW-10 and CUS50D sensors, with regression curve. Only data marked as “ $\times$ ” were used for the calibration. Grey bars indicate a possible range of optimal dosage of 38% based on previous work by (Shaw et al., 2022).

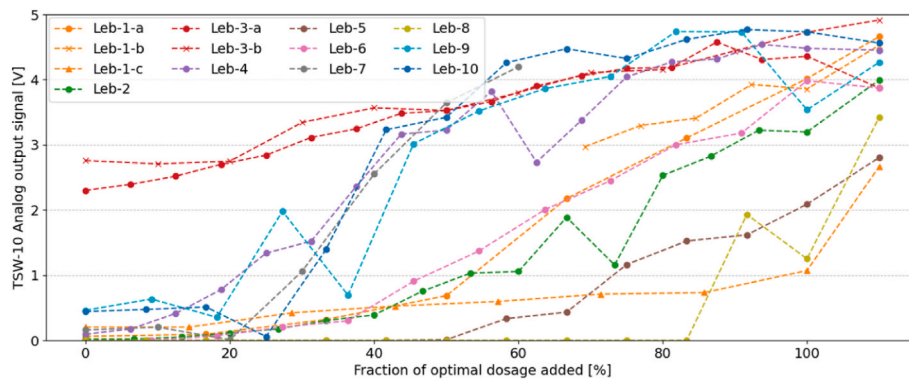


Fig. 4. Light-attenuation readings with TSW-10 during the stepwise addition of conditioner towards the dosage determined with jar tests (100%). Each point depicts the averaged voltage measurements from TSW-10 around dosage. A higher voltage reading on the y-axis means better solid-liquid separation. Replicates are indicated with letters after the sample name.

To validate these results, five additional trials were carried out, with step-wise addition of conditioners until the TSW-10 sensor output reached a value of 4.0 V. In other words, the decision to add another step-wise dose of conditioner was made after reading the output at each dosing step until an output of 4.0 V. For these trials, composites of Leb-1 to 10 were used, labelled as Auto-1 to 5 (dosage and TSW-10 output signals in Fig. S6). For three out of the five trials, the correct dosage was found in less than 3 min, which is substantially faster than conducting a jar test (up to 60 min) (Velkushanova et al., 2021), though two trials were not able to reach optimal dosage due to sensor fouling (Auto-5) and a programming error (Auto-3). Importantly, the reduction in TSS after geotextile treatment during these automated dosing trials (77% average) was in the same range as jar tests (83%) (Table 3). Thus, this is a proof of concept that indicates that a simple, low-cost light-attenuation sensor can be used to control conditioner dosing on a large scale. However, sensor fouling due to the stickiness of flocs needs to be considered when implementing this approach, for example with regular or automated cleaning of the sensor surface.

3.2.3. Effluent monitoring (CUS50D and TSW-10 light attenuation)

To evaluate their effectiveness in monitoring dewatering performance, CUS50D and TSW-10 sensors were installed to measure the effluent TSS after dewatering with a geotextile. Sensor readings of CUS50D and TSW-10 were plotted against TSS measured in the laboratory (Fig. 5). The calibration curve from Fig. 2 was added, with grey shading indicating a 10% range, to evaluate whether the same

calibration curve is valid also for lower TSS. In the resulting graphs, most of the data points lay in or just outside the 10% range, even though the calibration in Fig. 2 was conducted for a higher TSS range of 2.5–27 g/L. The only sample that deviated strongly from both curves was Leb-10; here, small flocs were visible, which led to a high heterogeneity in the light attenuation, disrupting the sensor measurement.

The previously discussed strong light attenuation by Leb-4, potentially caused by dark color and fine particles, could not be observed after adding conditioners. This indicates that conditioners can reduce differences in attenuation patterns (e.g. color and small particles) for more reliable attenuation measurements in the top layer of the tanks after flocculation, even when the influent wastewater displays differing characteristics. The CUS50D calibration curve, yet not the TSW-10 one, had two other distinct points, Leb-8 and Leb-9, that diverged from the anticipated trend. This might be due to a reduced precision at lower concentrations of solids, as the distance between the light emitter and receiver (5 mm) was optimized for higher concentrations of solids.

These measurements indicate that both sensors can be used for a wide range of TSS measurements, including for monitoring the effluent after dewatering. This is particularly relevant when downstream treatment processes can fail due to sudden high solids loadings. An example of this comes from Leb-1-c, as solid-liquid separation was low in this case (Table 4) compared to the expected (Leb-1-a and Leb-1-b), and the effluent reading of both sensors clearly indicated a high TSS concentration for Leb-1-c.

Overall, these measurements support to upscale from the pilot plant

Table 3

TSS and chemical oxygen demand (COD) measurements of stored wastewater (influent) and after conditioning and geo-textile (effluent).

Wastewater	Influent TSS [g/L]	Effluent TSS [g/L]	TSS reduction	Influent COD [mg/L]	Effluent COD [mg/L]	COD reduction
Leb-1-a	2.9	0.2	93%	3148	227	93%
Leb-1-b	2.9	0.5	83%	–	–	–
Leb-1-c	2.9	2.8	3%	–	–	–
Leb-2	5.8	0.2	96%	11,750	426	96%
Leb-3-a	1.2	0.7	45%	5052	1028	80%
Leb-3-b	1.2	0.6	50%	–	–	–
Leb-4	4.2	1.5	65%	6900	3005	56%
Leb-5	9.1	1.2	87%	12,450	1745	86%
Leb-6	1.1	0.0	96%	1676	119	93%
Leb-7	3.9	0.4	90%	4760	330	93%
Leb-8	15.7	0.2	99%	18,300	302	98%
Leb-9	1.9	0.2	92%	2272	458	80%
Leb-10	4.6	0.8	82%	2698	1445	46%
Auto-1	1.1	0.6	45%	1676	834	50%
Auto-2	6.0	0.8	87%	8093	3286	59%
Auto-3	6.1	2.2	64%	1394	1180	15%
Auto-4	1.2	0.0	97%	5052	153	97%
Auto-5	4.5	0.3	94%	6017	390	94%

Effluent monitoring (CUS50D and TSW-10 light attenuation).

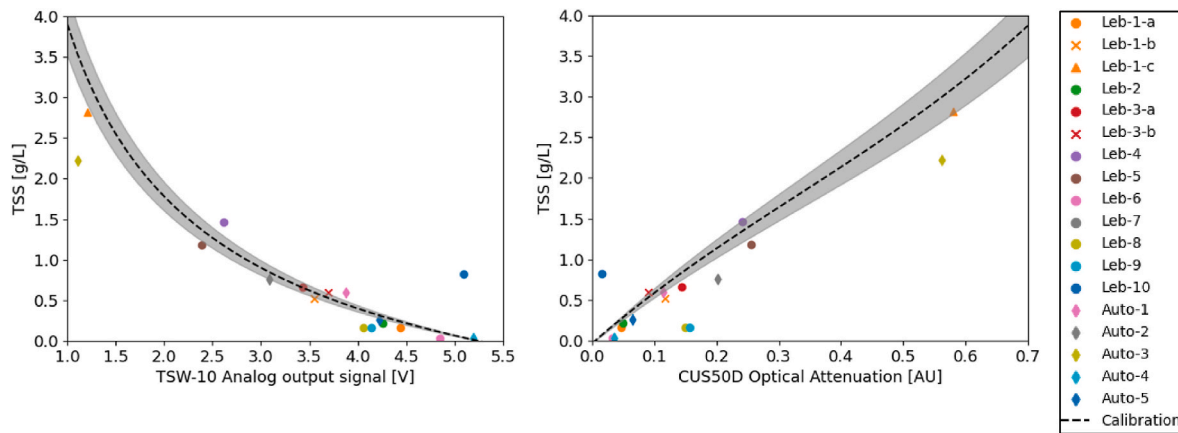


Fig. 5. Sensor readings from CUS50D and TSW-10 as compared to laboratory TSS measurements of the effluent of the dewatering pilot. Calibration curve is based on calibration in Fig. 2, and grey shading indicates 10% deviation from calibration curve.

Table 4

Summary of selection criteria for the evaluated methods to predict total suspended solids (TSS). x indicates prediction, with “+” use recommended in the trialed measurement range; “(+)” a limited accuracy over the trialed measurement range (0–25 gTSS/L); “-” use not recommended in the trialed measurement range.

Expected variability in	Coriolis-flow	Micro-wave	Light scattering	Light attenuation	Example
TSS	+	+	(+)	+	Blackwater from no-mix toilets (low urine contribution)
Salts	-	+ ^a	+	+	Blackwater from mix toilets (variable urine content)
Particle size and shape	-	(+)	(+)	(+)	Stored wastewater from unlined containments
FOG	-	-	+	+	Stored wastewater from restaurants

^a With electrical conductivity compensation.

to a full-scale fecal sludge treatment plant. In order to successfully scale, the next steps would be to determine the location of the measurement, meaning specifically the influent, the flocculation tank or the effluent. The actual characteristics of the wastewater and the measurement target would then guide the selection of the sensor (see next section ‘Implications’). Once selected, the sensor would need to be calibrated for the parameter of interest (e.g. TSS, TS, conditioner dosing). As a starting point, we suggest using the basic functions given in Table S2 (e.g. exponential function for TSW-10). It is important to calibrate the sensors in the actual wastewater of the implementation. In the case of stored wastewater (fecal sludge), several samples covering the expected parameter range must be collected for calibration. For each sample, a laboratory measurement and the sensor measurement would then be used to calibrate to the predictive model. This process must be repeated periodically during sensor operation to validate or refine the calibration.

4. Implications

The results of this study can guide the choice of sensor selection for measuring solids in applications for treatment of wastewater from non-

sewered sanitation systems with highly variable characteristics. However, based on the results, there is no one-size-fits-all sensor technology for this application. An indication of which sensor is expected to work best under different conditions is provided in Table 4. To facilitate initial sensor selection, the range of expected wastewater characteristics in a particular system should be first evaluated with laboratory measurements or estimated from the inflow wastewater sources.

There is a variety of examples where this is relevant. Micro-wave sensors might be the best option for measuring in a consistent wastewater composition with high TSS variability, such as treating blackwater from flush or low-flush toilets in buildings, neighborhoods or cruise ships. However, in smaller scale systems (e.g., RTTC Reinvent the toilet challenge), simpler and smaller sensors such as the optical sensors might be the better choice. Selection is more difficult if wastewater composition is less consistent. For example, if cooking wastewater is connected to the same system, sensors with a lower sensitivity to FOG (e.g., optical sensors) would be better, as cooking wastewater can contribute to a high variability of FOG. The same logic applies for semi-centralized treatment of stored wastewater at fecal sludge treatment plants. In this case, the inputs into the stored wastewater are often unknown, and can include a high variability of inorganic particles due to soil from unlined containments, salts due to industrial wastewater or high urine content (e.g., public toilets) and FOG from containments connected to restaurants. As such, a light-attenuation sensor might be a nice compromise, as it should better handle the unknown inflow, although it may potentially not cover the entire range of solids.

The application of these results is not limited to wastewater from non-sewered sanitation. The results can be transferred to other wastewater types where high variations of TSS are observed. For example, in the leachate of a landfill observed ranges are 3–14 gTSS/L (Abd El-Salam & I. Abu-Zuid, 2015) or 2–9 gTSS/L (Al-Wabel et al., 2011), in swinery wastewater 23–43 gTSS/L (Boursier et al., 2005) or in wastewater from pulp and paper mills 1–23 gTSS/L (Pokhrel and Viraraghavan, 2004). However, calibration curves from this study should not be directly applied to other wastewater types, as the calibration is strongly depended on the composition matrix of the wastewater type. Calibration must be done with samples from the wastewater type where the sensors will be used.

The next factor to consider is the required accuracy of the measurement in the process control or monitoring application, and especially the acceptable tolerance in accuracy. For example, some applications will only require an analysis of the trend of the measurement instead of an exact absolute value for process control, as illustrated by the example in this study of measuring light attenuation in the flocculation tank. Here, the prediction of conditioner dosing tolerated high measurement inaccuracies, indicating that it may be possible to use a

sensor even if its prediction accuracy is low (>10% deviation).

Additionally, sensor cost must be a factor. While highly engineered sensors like optical-light-scattering CUS51D outcompete low-cost sensors in user-friendliness (installation and data processing), material and measurement robustness, application range, and customer support, they are often designed to measure solids and turbidity in a broad range of application in drinking water and wastewater treatment, where high precision is needed also at lower turbidity (<100 formazine nephelometric units (FNU)) (Endress+Hauser, 2023; HACH, 2022). In contrast, wastewater from non-sewered sanitation often is in the range of 2000–10,000 FNU, in which case, the lower-cost light-attenuating sensors perform similarly well, as was shown in this study. This is particularly interesting, as light-attenuation sensors simply measure the amount of light that travels from a light source through the water to the detector. This technology can be built at a low cost (e.g. 35 USD by (Trevathan et al., 2022)) or is commercially available in low-cost versions (e.g., 100 USD with TSW-10 and microcontroller, as done in this study). However, when considering these low-cost options, worktime and material required for manufacturing and calibration should be considered in the cost calculations, as they can have similar overall costs as ready-to-use commercially available sensors (Trevathan et al., 2022).

Low-cost sensors are particularly important for small-scale treatment systems, where a higher number of sensors are needed per amount of wastewater treated compared to centralized systems. Here, sensor-based monitoring can reduce required labor costs, which are often higher than the sensor costs. The cost-efficiency can be increased by implementing sensors that cost less, or that most reliably fulfills the purpose. For example, the more reliable sensors are at predicting the failure of a system, the less frequent in-person inspection rounds are needed. Thus, in these small-scale systems, there is a break-even point in the number of correct failure predictions at which the use of sensors becomes cost-efficient (Schneider et al., 2023). Additionally, the treatment performance for small-scale treatment systems can be variable, which would be difficult to stay on top of with manual sampling. Sensors could more readily capture this variability and could prevent the release of contaminated water into the environment (Reynaert et al., 2024).

For larger scale treatment of stored wastewater (e.g., fecal sludge treatment plants), a high variability of wastewater characteristics is expected. Before implementing sensors for conditioner dosing, the cost-efficiency needs to be evaluated based on full-scale data that includes investment costs, sensor maintenance, conditioner savings, and labor savings compared to a fixed conditioner dosage or manual jar testing.

5. Conclusions

The conclusions of this study are.

- This study is a proof-of-concept which demonstrates that sensors can be used to predict TSS in wastewater from non-sewered sanitation, which can be applied in process control, including automating dosing for chemically enhanced dewatering.
- Optical light-attenuation sensors predicted TSS with the overall lowest error and have the potential to be built locally if the required electrical components are available.
- Simple, low-cost light-attenuation sensor can be used to automate conditioner dosing on a larger scale, provided that the dosing is controlled based on changes in the sensor signal rather than relying on an exact TSS prediction value.
- Sensors need to be selected based on inflow characteristics specific to the local context and the measurement application, as there is potentially no sensor that can be calibrated to predict TSS over the entire range of characteristics.
- Sensor manufacturers can support this process by adapting and developing sensor technology for specific characteristics of TSS, VSS/TSS, FOG, and EC observed in non-sewered sanitation, including validation and case study reports.

CRedit authorship contribution statement

Michael Vogel: Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Maxwell Bergström:** Writing – review & editing, Investigation, Data curation. **Rita Al JahJah:** Writing – review & editing, Investigation. **Charbel Hankache:** Writing – review & editing, Investigation. **Rabea Touma:** Writing – review & editing, Investigation. **Linda Strande:** Writing – original draft, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Endress+Hauser (Schweiz) AG donated Teqwave F and Promass Q300 and consulted on the TSS sensorpilot construction. CUBEX MENA built and supported the operation of the dewatering pilot.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jenvman.2024.123554>.

Data availability

Data will be made available on request.

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